

## Lecture 2: Vectors in $\mathbb{R}^3$

### Background & Motivation

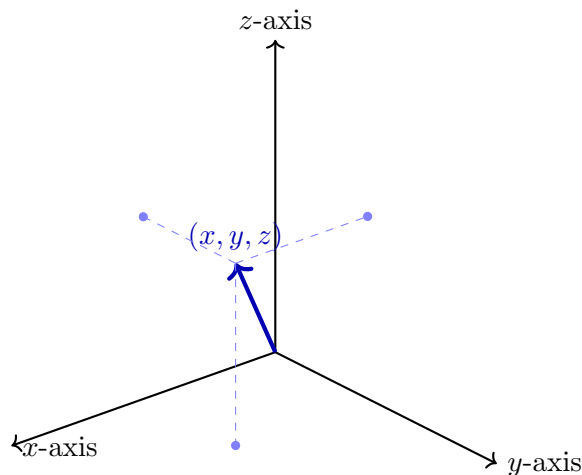
Last time we set up 3D coordinates. Now we need a language to describe direction and magnitude in space—that's what vectors give us. Vectors are the building blocks of everything in this course: lines, planes, curves, derivatives, integrals, and fields all use vector notation.

### 1. Vectors — Definitions

#### Definition 1: Vector

A **vector** is a quantity carrying both **magnitude** and **direction** — visually, an arrow. Two arrows are considered the *same* vector whenever they have the same length and direction, no matter where they are drawn, so a vector is really an equivalence class of parallel arrows. Placing its tail at the origin, the tip lands at a specific point  $(v_1, v_2, v_3)$ , and those coordinates are the **components** of  $\mathbf{v}$ .

- Written  $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ ; angle brackets distinguish vectors from points  $(a, b, c)$



A vector in  $\mathbb{R}^3$  represented by its components along the  $x$ ,  $y$ , and  $z$  axes. Dashed lines mark the projections onto the three coordinate planes.

#### Definition 2: Magnitude (Norm)

The **magnitude** (or **norm**)  $\|\mathbf{v}\|$  is the length of the vector — the Euclidean distance from its tail to its tip. When  $\mathbf{v}$  is drawn with its tail at the origin,  $\|\mathbf{v}\|$  is literally the distance from the origin to

the terminal point, and it is unchanged by rotating or translating the coordinate system. In short, magnitude strips away direction and records only *how much* of the vector there is. It is computed by the Pythagorean theorem applied to the components:

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

### Definition 3: Unit Vector

A vector with magnitude **1**. To **normalize**:

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$$

Standard unit vectors:  $\hat{\mathbf{i}} = \langle 1, 0, 0 \rangle$ ,  $\hat{\mathbf{j}} = \langle 0, 1, 0 \rangle$ ,  $\hat{\mathbf{k}} = \langle 0, 0, 1 \rangle$

### Definition 4: Direction Cosines

For  $\mathbf{v} = \langle a, b, c \rangle$ , the angles  $\alpha, \beta, \gamma$  with the  $x$ -,  $y$ -,  $z$ -axes:

$$\cos \alpha = \frac{a}{\|\mathbf{v}\|}, \quad \cos \beta = \frac{b}{\|\mathbf{v}\|}, \quad \cos \gamma = \frac{c}{\|\mathbf{v}\|}$$

Key identity:  $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

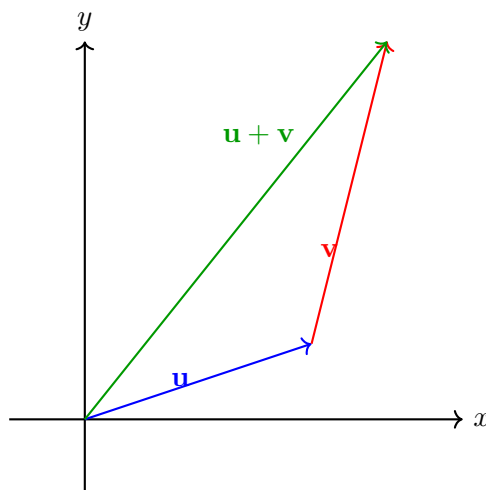
### Vector Arithmetic

If  $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$  and  $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ :

$$\mathbf{u} + \mathbf{v} = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle$$

$$c\mathbf{u} = \langle cu_1, cu_2, cu_3 \rangle$$

– Addition is “tip-to-tail”; scalar multiplication scales length



Tip-to-tail method of vector addition: placing  $\mathbf{v}$  at the tip of  $\mathbf{u}$  results in  $\mathbf{u} + \mathbf{v}$ .

## 2. The Dot Product

### Definition 5: Dot Product

Component formula:

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3$$

Geometric formula:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

– The dot product is a **scalar** (number, not a vector)

### Dot Product Facts

- $\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$
- Angle between vectors:  $\theta = \arccos\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}\right)$

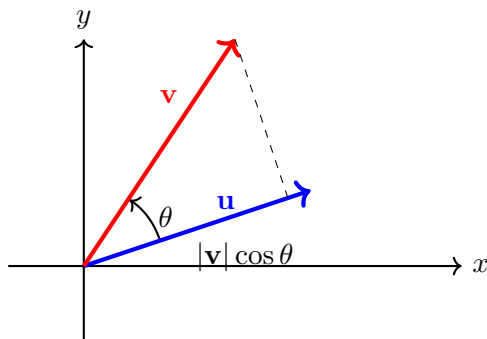
### Definition 6: Vector Projection

The **projection** of  $\mathbf{u}$  onto  $\mathbf{v}$  is the “shadow” that  $\mathbf{u}$  casts along the line of  $\mathbf{v}$  — the part of  $\mathbf{u}$  that points in the direction of  $\mathbf{v}$ . It is a vector parallel to  $\mathbf{v}$  whose length is the *scalar component*  $\text{comp}_{\mathbf{v}} \mathbf{u} = \|\mathbf{u}\| \cos \theta$ .

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \mathbf{v}$$

Scalar component:

$$\text{comp}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|}$$



The dot product interpreted as the projection of  $\mathbf{v}$  onto  $\mathbf{u}$ , scaled by  $\cos \theta$ .

### Example 1: Crane Cable Forces

Two crane cables exert forces  $\mathbf{F}_1 = \langle 120, 0, 80 \rangle$  N and  $\mathbf{F}_2 = \langle -50, 75, 60 \rangle$  N.

- Find the total force  $\mathbf{F}_1 + \mathbf{F}_2$ .
- Find a unit vector in the direction of the total force.
- What force  $\mathbf{F}_3$  would perfectly counterbalance  $\mathbf{F}_1 + \mathbf{F}_2$ ?

1. Total force:

$$\mathbf{F}_1 + \mathbf{F}_2 = \langle 70, 75, 140 \rangle \text{ N}$$

2. Magnitude:

$$\|\mathbf{F}_1 + \mathbf{F}_2\| = \sqrt{70^2 + 75^2 + 140^2} = \sqrt{4900 + 5625 + 19600} = \sqrt{30125} \approx 173.6 \text{ N}$$

3. Unit vector:

$$\hat{\mathbf{F}} = \frac{1}{173.6} \langle 70, 75, 140 \rangle \approx \langle 0.403, 0.432, 0.807 \rangle$$

4. Counterbalance:  $\mathbf{F}_3 = -(\mathbf{F}_1 + \mathbf{F}_2) = \langle -70, -75, -140 \rangle \text{ N}$

### Example 2: Dot Product and Angle

Let  $\mathbf{a} = \langle 3, 0, 4 \rangle$  and  $\mathbf{b} = \langle 1, 2, -1 \rangle$ . Find  $\mathbf{a} \cdot \mathbf{b}$  and the angle between them.

1. Dot product:

$$\mathbf{a} \cdot \mathbf{b} = (3)(1) + (0)(2) + (4)(-1) = 3 + 0 - 4 = -1$$

2. Magnitudes:

$$\|\mathbf{a}\| = \sqrt{9 + 0 + 16} = 5, \quad \|\mathbf{b}\| = \sqrt{1 + 4 + 1} = \sqrt{6}$$

3. Angle:

$$\cos \theta = \frac{-1}{5\sqrt{6}} \implies \theta = \arccos\left(\frac{-1}{5\sqrt{6}}\right) \approx 94.7^\circ$$

Since  $\theta > 90^\circ$ , the vectors point in “generally opposite” directions.

### Example 3: Projection

Project  $\mathbf{a} = \langle 3, 4, 5 \rangle$  onto  $\mathbf{n} = \langle 1, 1, 1 \rangle$ .

1. Compute pieces:

$$\mathbf{a} \cdot \mathbf{n} = 3 + 4 + 5 = 12, \quad \|\mathbf{n}\|^2 = 3$$

2. Projection:

$$\text{proj}_{\mathbf{n}} \mathbf{a} = \frac{12}{3} \langle 1, 1, 1 \rangle = \langle 4, 4, 4 \rangle$$

3. Perpendicular component:  $\mathbf{a} - \text{proj}_{\mathbf{n}} \mathbf{a} = \langle 3, 4, 5 \rangle - \langle 4, 4, 4 \rangle = \langle -1, 0, 1 \rangle$

## 3. The Cross Product

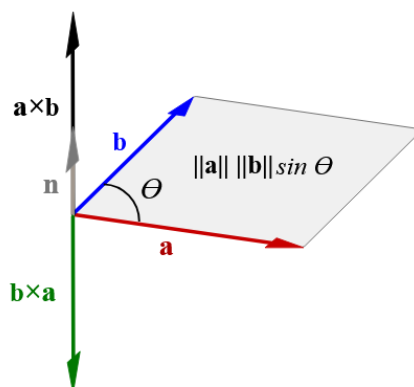
### Definition 7: Cross Product

Determinant formula:

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \langle u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1 \rangle$$

Properties:

- $\mathbf{u} \times \mathbf{v}$  is **perpendicular to both  $\mathbf{u}$  and  $\mathbf{v}$**
- $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta$  – area of parallelogram
- Direction via the **right-hand rule**
- $\mathbf{u} \times \mathbf{v} = -\mathbf{v} \times \mathbf{u}$  (anti-commutative)



Cross product:  $\mathbf{a} \times \mathbf{b}$  is perpendicular to both, with magnitude =  $\|\mathbf{a}\| \|\mathbf{b}\| \sin \theta$  (parallelogram area).

### Cross Product Facts

- $\mathbf{u} \parallel \mathbf{v} \iff \mathbf{u} \times \mathbf{v} = \mathbf{0}$
- $\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}, \quad \hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}, \quad \hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}$
- $\mathbf{u} \times \mathbf{u} = \mathbf{0}$  always

### Example 4: Cross Product Computation

Compute  $\mathbf{a} \times \mathbf{b}$  for  $\mathbf{a} = \langle 3, 0, 4 \rangle$  and  $\mathbf{b} = \langle 1, 2, -1 \rangle$ .

1. Set up the determinant:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 3 & 0 & 4 \\ 1 & 2 & -1 \end{vmatrix}$$

2. Expand:

$$= \hat{\mathbf{i}}(0 \cdot (-1) - 4 \cdot 2) - \hat{\mathbf{j}}(3 \cdot (-1) - 4 \cdot 1) + \hat{\mathbf{k}}(3 \cdot 2 - 0 \cdot 1) = \langle -8, 7, 6 \rangle$$

3. Verify orthogonality:  $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = (3)(-8) + (0)(7) + (4)(6) = -24 + 0 + 24 = 0 \quad \checkmark$

## 4. Scalar Triple Product

### Scalar Triple Product

$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$  – gives the **volume of the parallelepiped** spanned by  $\mathbf{a}, \mathbf{b}, \mathbf{c}$

$$\text{Volume} = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$$

### Example 5: Scalar Triple Product — Volume

Find the volume of the parallelepiped spanned by  $\mathbf{a} = \langle 1, 0, 2 \rangle$ ,  $\mathbf{b} = \langle 3, 1, 0 \rangle$ ,  $\mathbf{c} = \langle 0, 2, 1 \rangle$ .

1. Cross product  $\mathbf{b} \times \mathbf{c}$ :

$$\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 3 & 1 & 0 \\ 0 & 2 & 1 \end{vmatrix} = \langle 1, -3, 6 \rangle$$

2. Triple product:

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (1)(1) + (0)(-3) + (2)(6) = 13$$

3. Result: Volume =  $|13| = \boxed{13}$  cubic units.

*Show all work for full credit.*

## Practice Problems

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**Problem 1.** Let  $\mathbf{v} = \langle 3, 4, 12 \rangle$ . Find  $\|\mathbf{v}\|$ , the unit vector  $\hat{\mathbf{v}}$ , and the direction cosines.

Answer:  $\|\mathbf{v}\| = 13$ ,  $\hat{\mathbf{v}} = \langle 3/13, 4/13, 12/13 \rangle$

Direction cosines:  $\cos \alpha = 3/13$ ,  $\cos \beta = 4/13$ ,  $\cos \gamma = 12/13$

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**Problem 2.** Compute  $\mathbf{u} \cdot \mathbf{v}$  and  $\mathbf{u} \times \mathbf{v}$  for  $\mathbf{u} = \langle 2, -1, 3 \rangle$  and  $\mathbf{v} = \langle 4, 0, -2 \rangle$ .

Answer:  $\mathbf{u} \cdot \mathbf{v} = 2$ ,  $\mathbf{u} \times \mathbf{v} = \langle 2, 16, 4 \rangle$

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**Problem 3.** Find the angle between  $\mathbf{u} = \langle 1, 1, 0 \rangle$  and  $\mathbf{v} = \langle 0, 1, 1 \rangle$ .

Answer:  $\theta = \pi/3 = 60^\circ$

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**Problem 4.** Find  $\text{proj}_{\mathbf{n}} \mathbf{a}$  where  $\mathbf{a} = \langle 3, 4, 5 \rangle$  and  $\mathbf{n} = \langle 1, 1, 1 \rangle$ . Also find the perpendicular component.

Answer:  $\text{proj}_{\mathbf{n}} \mathbf{a} = \langle 4, 4, 4 \rangle$ , perpendicular component =  $\langle -1, 0, 1 \rangle$

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**Problem 5.** Three points  $P_1 = (1, 2, 3)$ ,  $P_2 = (4, -1, 2)$ ,  $P_3 = (-2, 3, 5)$  define a plane. Use the cross product to find a normal vector.

Answer:  $\mathbf{n} = \langle -5, -3, -6 \rangle$

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**Problem 6.** Find the volume of the parallelepiped spanned by  $\mathbf{a} = \langle 2, 1, 0 \rangle$ ,  $\mathbf{b} = \langle 1, -1, 3 \rangle$ ,  $\mathbf{c} = \langle 0, 2, 1 \rangle$ .

Answer:  $\text{Volume} = 13$